

Fall 2024
CS301
Assignment No. 01

```
#include <iostream>

class Product {
private:
    int data;
    Product* next;
    int productId;

public:
    // Constructor
    Product() : data(0), next(nullptr), productId(0) {}

    // Getter for data
    int get() {
        return data;
    }

    // Setter for data
    void set(int value) {
        data = value;
    }

    // Getter for next
    Product* getNext() {
        return next;
    }

    // Setter for next
    void setNext(Product* nextNode) {
        next = nextNode;
    }

    // Getter for productId
    int getProductId() {
        return productId;
    }

    // Setter for productId
    void setProductId(int pid) {
        productId = pid;
    }
};

class Stack {
private:
    Product* head;
```

```

public:
    // Constructor
    Stack() : head(nullptr) {}

    // Push function to add a product to the stack
    void push(int x) {
        Product* newProduct = new Product();
        newProduct->set(x);
        newProduct->setNext(head); // Point new product to the current head
        head = newProduct; // Update head to new product
    }

    // Function to get the top product's data
    int top() {
        if (head != nullptr) {
            return head->get();
        }
        throw std::runtime_error("Stack is empty");
    }

    // Pop function to remove a product from the stack
    int pop() {
        if (head == nullptr) {
            throw std::runtime_error("Stack is empty");
        }
        Product* temp = head;
        int data = head->get();
        head = head->getNext(); // Move head to the next product
        delete temp; // Free memory
        return data;
    }

    // Destructor to clean up the stack
    ~Stack() {
        while (head != nullptr) {
            pop(); // Pop all elements
        }
    }
};

int main() {
    Stack productStack;

    // Set productId based on example logic
    int studentIdLastTwoDigits = 70; // Replace with your actual last two digits
    int productId = 2400 + studentIdLastTwoDigits; // 24 + last two digits
    std::cout << "Product ID to be set: " << productId << std::endl;

    // Create and push products onto the stack
    for (int i = 1; i <= 5; i++) { // Example to push 5 products
        productStack.push(i * 10); // Pushing some data (10, 20, 30, ...)
    }

    // Display the top product data
    std::cout << "Top product data: " << productStack.top() << std::endl;

```

```
// Pop products and display their data
while (true) {
    try {
        std::cout << "Popped product data: " << productStack.pop() << std::endl;
    } catch (std::runtime_error& e) {
        std::cout << e.what() << std::endl;
        break; // Exit loop when stack is empty
    }
}

return 0;
}
```